

Pranesh S

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SUMMARY

AI/ML Engineer building production **LLM systems** in **Python** — agentic pipelines on **MCP** and **AWS Bedrock AgentCore**, **RAG** platforms, and multi-agent workflows. **Immediate Joiner**. Open-source maintainer: **100+ public repos**, **270+ stars**, **3 PyPI packages**, **35+ deployed AI demos**.

EXPERIENCE

- AI Engineering Intern — Wipro Connected Services (formerly Harman DTS)** Feb 2026 – Jun 2026
- Achieved **82% executable-SQL generation (37/45)** on the **Spider benchmark** by building an **NL2SQL2NL** system on the **Model Context Protocol (MCP)** with schema-aware prompting, enabling context-aware, multi-turn plain-English querying of relational databases for business users.
 - Reduced SQL composition time by **40%** and passed **100% of an 8-query business demo suite** by delivering the pipeline to non-technical stakeholders with conversational, **Matplotlib**-based chart generation.
 - Deployed team-built agents and **MCP servers** to **AWS Bedrock AgentCore Runtime** by containerizing **FastAPI** services as **linux/arm64 Docker** images against the AgentCore service contract and publishing to **Amazon ECR**, covering both HTTP and MCP runtime protocols.
 - Achieved **92% tool-agent alignment accuracy** and **35% faster agent development** by building a **Multi-Agent Workflow Optimizer** with ontology-driven requirement parsing and **AgentBank** validation, preventing **5+ duplicate tool implementations**.

PROJECTS

- turnloop — Agentic Coding Harness** | *Python, Textual, vLLM, Modal, PyPI* github.com/Pranesh-2005/turnloop
- Cut agent **input-token usage by 20%** and tool-call error rate to **0%** by building an **evaluation harness** and running a **7-arm, 84-run ablation** on agentic-harness design; disabling context compaction cost **39% more input tokens** (273K vs 196K median).
 - Shipped the harness to **PyPI** — **21,300 lines of Python**, **383 tests**, and only **5 runtime dependencies**, no vendor SDKs — and self-hosted the **744B-parameter GLM-5.2 MoE** evaluation target on **vLLM** across **4x H200** GPUs via **Modal**.
- CloudLens — Multi-Agent AWS Operations Platform** | *LangGraph, FastAPI, A2A* github.com/Pranesh-2005/Cloudlens
- Built a **LangGraph** supervisor routing to **6 specialist agents** for cost, forecasting, security, and deployment across tenant AWS accounts, exposed through both an **A2A protocol** server and a **FastMCP** endpoint, with **Fernet**-encrypted credentials and human approval gating on every mutating action.
- RagObserve — RAG Observability Platform** | *Python, FastAPI, SQLite, PyPI* github.com/Pranesh-2005/RagObserve
- Cut tracing overhead **136x** (379.8 ms to 2.8 ms per trace) by moving the SQLite writer to **WAL** journaling, making the tracer safe on an application's hot path; covers **11 LLM providers** with **LangChain** and **LlamaIndex** adapters, fully offline.
- LLM Chatbot with Long-Term RAG Memory** | *Qwen2.5-7B, LoRA, Unsloth, Modal, GGUF*
- Fine-tuned **Qwen2.5-7B-Instruct** with **LoRA (Unsloth)** on serverless **Modal A10G** GPUs using **14K instruction-tuning samples** engineered from **120K+ raw chat messages**; added long-term memory via **RAG** retrieval over **30K+ embedded bursts** and exported **4-bit GGUF** models for local inference.

TECHNICAL SKILLS

Languages & Backend: Python, Java, SQL, Data Structures & Algorithms, System Design, FastAPI, Flask, REST APIs, PostgreSQL, ChromaDB

Generative AI & Agents: AI Agents, Agentic AI, Multi-Agent Systems, MCP, A2A, LangChain, LangGraph, CrewAI, RAG, embeddings & semantic search, LLM fine-tuning (LoRA, QLoRA), prompt engineering, Context Engineering, LLM evaluation & benchmarking

Cloud & MLOps: AWS Bedrock AgentCore, Amazon ECR, Azure AI Foundry, Modal, Docker, GitHub Actions, CI/CD, Git/GitHub (Version Control), Hugging Face, PyPI, Vercel, Render, Supabase

Machine Learning & Deep Learning: Transformers, PyTorch, TensorFlow, NumPy, Pandas, scikit-learn, Deep Learning, NLP, model evaluation

ACHIEVEMENTS & OPEN SOURCE

- github-readme-stats-fast:** GitHub profile stats generator with a companion GitHub Action — **270+ stars**, **59 forks**
- PyPI packages:** *turnloop* (agentic coding CLI), *guardix* (prompt-injection guard), *Text-Emotion*, *DocTok*
- Built and deployed **35+ AI demos** on Hugging Face spanning RAG, agents, and model-internals visualizers (GPT-2, ViT, YOLO)

EDUCATION

Bannari Amman Institute of Technology
Bachelor of Technology (B.Tech) in Artificial Intelligence and Machine Learning — CGPA: 7.97/10

2022 – Apr 2026

CERTIFICATIONS

- Anthropic:** Introduction to Model Context Protocol; Model Context Protocol: Advanced Topics
- Microsoft:** Azure AI Fundamentals (AI-901)
- freeCodeCamp:** Machine Learning with Python